

Biography

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One of my earliest memories is doing simple algebra problems on the back of napkins and envelopes with my dad. These exercises instilled a love for math in me early on, and throughout grade school it was probably my favorite subject. Somewhere along the way I also became interested in computer science, inspired by my dad's job as a computational linguist, and then developed an affinity for AI as well. Throughout high school, I became more and more involved in AI/ML classes and clubs, eventually completing a year-long research project my senior year on reinforcement learning (a subfield of ML). Entering CMU I retained my long-time interest in math, so I decided I would double major in math and something ML-related. I wound up as a math major with an ML minor, but as time went on, I became more interested in math and less interested in ML. The schedule of classes then worked out this semester that I even had to drop the ML minor and fully commit to math. With the ML minor gone, I thought I was basically done with that part of my life.

This semester, however, I reached out to Professor Florian Frick in the math department for research project suggestions, and to my surprise he suggested a project he called "Topology meets ML." I've grown quite fond of topology through my studies as a math major, so this project immediately piqued my interest, especially in combination with my background in ML. This then developed into the proposal I'm now submitting for SURF, which excites me because I get to combine two interests of mine that I've scarcely seen connected before. I think my background rather uniquely situates me to have good insight and make progress in this area, so I look forward to this opportunity.

Beyond project specifics, I am tentatively planning to pursue a PhD in math after I graduate from CMU so that I can do some sort of research-based job (academia, government labs, etc) for my career. To that end, I am looking to get a feel for what research looks like before I commit to doing it for five or so years, hence my participation in SURA last year and (hopefully) SURF this year.

Outside my academic work, I am very active in the Christian community at CMU, being a leader in both InterVarsity Christian Fellowship, a national college campus ministry, and Joyful Noise, CMU's Christian acappella group (which also happens to be the oldest acappella group at CMU). My involvement in the latter was inspired by my playing trumpet the last ten years, a hobby which I hope to get back into after college. I also like to cook, and recently I've been perfecting my steak technique (my girlfriend is now very fond of my steak, claiming it's the best she's ever had). Growing up I was always fond of playing video games with friends, though in recent years I've gravitated more to single player, story-focused games like *Hollow Knight* and its long-awaited sequel *Hollow Knight: Silksong* (both of which are absolute masterpieces, in my humble opinion).